



Financial Economic Dynamics
EC3108
Module Outline 26/27

Aims & Objective	Time is money — yet in a field built on that slogan, remarkably few of its models contain actual time!		
	Students arrive at this course already knowing a great deal of financial economics and knowing all through timeless models. The efficient frontier is a curve, not a path; the CAPM is a cross-sectional relation holding at a single instant; the present value is a formula, not a process; risk is a number, σ , sitting in a covariance matrix that does not move. The power of the static framework comes from what it leaves out: it delivers an optimum rather than a <i>trajectory</i> .		
	This course introduces time, or dynamics, in Financial Economics and asks what impact it has on our understanding of agents' financial decisions.		
	The course is organised into three blocks , each telling the same story from a different angle. A financial agent faces both a <i>personal decision</i> , her patience and risk aversion, and an <i>objective one</i> , the best investment or optimal risky bundle. Markets let her trade freely at given prices, separating these decisions: she can solve the objective problem first, ignoring her own preferences, and then use trading to reshape that optimal position into one that fits her personality.		
	Block 1 tells this story holding time fixed at one period, from Fisher's separation theorem through Tobin, Markowitz, and the CAPM. Block 2 tells the identical story once time is allowed to run, replacing single first-order conditions with Euler equations and Hamiltonians, and asking exactly when the dynamic answer collapses back to the static one and when it does not. Block 3 tells the story a third time, but now from the standpoint of whether agents can ever possess the kind of knowledge about future evolution of the economy that is assumed in the first two blocks. Once that assumption is amended, the analysis shifts: we ask what a portfolio, a price, or an entire financial system looks like without reliable knowledge of how real activity feeds back into financial valuations. Block 3 therefore exposes the limits of optimisation under perfect information and shows how financial systems behave when real-side dynamics are only partially understood. The three interconnected blocks of the course are designed to equip students to analyse financial systems dynamically and to understand financial decisions as outcomes of evolving real–financial interactions. Rather than treating these systems as linear or self-contained, the course fosters the ability to see financial and real spheres as mutually shaping components of a complex, adaptive economy. The dynamical systems method also introduces students to the stock–flow consistent analytical framework to the study of real–financial dynamics, an approach that is rarely covered in standard financial economics courses.		
Lecturer	Name Dr. Srinivas Raghavendra	Office 307, Cairnes Building	E-mail s.raghav@universityofgalway.ie
Overall Module Learning Outcomes	- Derive the static results of finance: Fisher's Separation Theorem, Tobin's two-fund separation, the Mean-Variance frontier, and the CAPM, and evaluate what each assumes about time and about the agent's knowledge. - Analyse the financial optimization problem as a dynamic sequential problem (discrete time) or an optimal-control problem (continuous time).		

	• Solve and analyse ordinary difference and differential equations through analytical and quantitative methods • Derive a dynamic optimum via the Euler equation or the Hamiltonian, and its costate variable, and analyse the resulting system's stability using phase diagrams, eigenvalues, and the trace–determinant plane • Analyse historical episodes using the three blocks' frameworks and examine the underlying dynamical mechanisms that may have driven the observed phenomena in those episodes.																
Graduate Attributes	Engaged students have opportunities through the coursework and the broader student experience to develop a wide range of knowledge, skills and dispositions. Upon completion of this module, you will develop the following key graduate attributes: • Knowledge: Logical and Analytical reasoning and Critical evaluation • Skills: Problem solving, creativity and communication • Disposition: Lifelong learning and personal effectiveness The graduate attributes are examined through various forms of assessments (see assessment section).																
Programme(s)	4FM2 (Financial Mathematics and Economics), Erasmus, and Visiting Students																
Delivery Format	This module will be delivered on campus. The module comprises 12 weeks of in-person lectures. Students are expected to attend and engage with class activities where appropriate. Classes are scheduled as follows: <table border="1"> <thead> <tr> <th>Day</th><th>Monday</th></tr> </thead> <tbody> <tr> <td>Time</td><td>10:00 – 11:00</td></tr> <tr> <td>Building</td><td>Cairnes</td></tr> <tr> <td>Room</td><td>CA001</td></tr> </tbody> </table> <table border="1"> <thead> <tr> <th>Day</th><th>Tuesday</th></tr> </thead> <tbody> <tr> <td>Time</td><td>15:00 – 17:00</td></tr> <tr> <td>Building</td><td>Lifecourse Institute</td></tr> <tr> <td>Room</td><td>LCI-G006 Seminar Room 5</td></tr> </tbody> </table> Map of University of Galway Campus and location of buildings is available here	Day	Monday	Time	10:00 – 11:00	Building	Cairnes	Room	CA001	Day	Tuesday	Time	15:00 – 17:00	Building	Lifecourse Institute	Room	LCI-G006 Seminar Room 5
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Attendance	Monitoring attendance has a positive impact on student engagement, especially when you can review your own attendance on Blackboard and receive emails about absence. You should attend in-person on-campus for all aspects of this module unless for legitimate reasons (for example, disability, medical, or family emergency) which you have reported to the lecturer. To support students, the J.E. Cairnes School of Business & Economics will monitor attendance at classes to (i) identify students that may be failing to engage with the programme and (ii) student participation in various modules. Quickly Attendance will be used within Blackboard to allow lecturers to take attendance at the module level, and for students to keep track of the same. Lectures will outline instructions for attendance in class. Student principles on general attendance can be found here																
Core Text	The module offers a new approach to teaching Financial Economics. I will have my notes for most of the units and suggest supplementary readings through the term. I list a few textbooks for refreshing foundational concepts in Financial economics and Dynamical Systems. <u>Financial Economics</u> 1. Bailey, R. E. (2005). <i>The Economics of Financial Markets</i>. Cambridge University Press.																

	2. Copeland, T. E., Weston, J. F., & Shastri, K. (2005). <i>Financial Theory and Corporate Policy</i> (4th ed.). Pearson Addison-Wesley. <u>Dynamical Systems</u> 1. Gandolfo, G. (1971). <i>Economic dynamics: Methods and models</i> (Vol. 16). Elsevier. 2. Shone, R. (2002). <i>Economic Dynamics</i>, 2nd edn. Cambridge University Press. 3. Raghavendra, S., & Piiroinen, P. T. (2023). <i>An introduction to economic dynamics: modelling, analysis and simulation</i>. Taylor & Francis. <u>Simulation platform:</u> https://raghavsriniv.github.io/econdyn/												
Assessment	End of Year Examination: 60% Continuous Assignment: 40% Details on the continuous assignments is given below and further details of the modes operandi will be provided during the term. The assessment are carefully designed to examine graduate attributes as follows: <table><tr><td></td><td>Graduate Attribute</td><td>Assessment</td></tr><tr><td>Knowledge</td><td>Logical and Analytical reasoning and Critical evaluation</td><td>CA and End Semester exam</td></tr><tr><td>Skills</td><td>Problem solving, creativity and communication</td><td>CA and End Semester exam</td></tr><tr><td>Disposition</td><td>Lifelong learning and personal effectiveness</td><td>CA</td></tr></table> Format of the end semester examination: 2-hour sit-in exam Format of Continuous Assessments: <i>There will two CAs, each worth 20%, after each Block (see the syllabus below) is delivered. The CAs will be in-class sit-in test for two hours.</i>		Graduate Attribute	Assessment	Knowledge	Logical and Analytical reasoning and Critical evaluation	CA and End Semester exam	Skills	Problem solving, creativity and communication	CA and End Semester exam	Disposition	Lifelong learning and personal effectiveness	CA
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Workload	<table><tr><td>Credit weighting</td><td>x ECTS [Details on ECTS and workloads available here]</td></tr><tr><td>Online weekly instructional hours</td><td>5 ECTS</td></tr><tr><td>Independent assignment work (hours)</td><td>24</td></tr><tr><td>Independent study (hours)</td><td>12</td></tr><tr><td>Associated hours</td><td>89</td></tr><tr><td>Total Student Effort</td><td>125</td></tr></table>	Credit weighting	x ECTS [Details on ECTS and workloads available here]	Online weekly instructional hours	5 ECTS	Independent assignment work (hours)	24	Independent study (hours)	12	Associated hours	89	Total Student Effort	125
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Plagiarism	Plagiarism is the use of another person’s ideas or work without appropriate acknowledgement or credit. Plagiarism may be intentional or unintentional. If you are unsure of the acceptable methods of acknowledgment you should refer to the University’s Code of Practice for Dealing with Plagiarism, consult with your lecturer or the library staff. Proven plagiarism is a very serious matter which may result in severe disciplinary action and/or exclusion from the University. Ensure all assignment submissions include a signed plagiarism statement.												
Referencing & Citation	Correct referencing and citation avoids plagiarism. There are varying referencing styles available but the most popular is the Harvard Referencing Style. Details on												

	how to reference journal articles, books, electronic information and various other supports is available from the University of Galway Library at the following link: https://libguides.library.universityofgalway.ie/StartingyourResearchatUniversityofGalwayLibrary/Referencing/Citation
LENS Report	Support is available to students with a disability and is granted on the basis of medical evidence. Where applicable, a LENS Report (Learning Educational Needs Summary) should be completed and the relevant disciplines notified. Please ensure you are registered with the Disability Support Service and have a LENS Report.
Classroom Charter	We are guided by our core values of respect in our actions, excellence in our work, openness in our outlook and sustainability in our approach. We focus on equality by treating people with respect and dignity, and providing everyone with access and opportunities to fulfil their potential. We embrace diversity to enrich our School environment by sparking innovation and creating a rewarding and vibrant student community. Within the J.E. Cairnes School of Business & Economics, at a module-level we aim to create and nurture an environment that is respectful and inclusive for all. To achieve this, we will: • Treat each other with dignity and respect. • Listen to each other's point of view, recognising that there may be disagreement. • Keep discussion, comments, and module feedback on the topic, and off the people. • Do not use inflammatory or offensive language, sarcasm or raised voices.

Schedule of Module Content & Learning Material

Module overview: The Three Blocks at a Glance

Wks	Block	The question	What happens
1–4	1. Mainstream Static	What is the optimal financial decision, holding time fixed at one period?	Fisher's separation theorem → Tobin's two-fund separation → Markowitz's mean-variance frontier → the CAPM as market equilibrium
5–8	2. Mainstream Dynamic	What changes when the same decisions are made repeatedly, or continuously, through time?	The dynamical-systems toolkit (no dynamic programming) → Samuelson's myopia theorem → Merton's continuous-time portfolio problem and hedging demand → dynamic asset pricing and volatility as a process
9–11	3. Heterodox	What does each answer look like once an agent cannot know what the mainstream machinery assumes she knows?	Keynes and Kalecki on liquidity preference and increasing risk → Minsky and Keen on financial instability and debt dynamics → heterogeneous agents, reflexivity, and the three crises
12	Review & Assessment	What has each block let us see, and what questions each made it impossible to ask?	The three blocks reviewed – what they focus and miss out; course and teaching evaluation

Syllabus

	Units	Readings / Activities
Block 1: Mainstream Static	Week 1: Individual optimization under Certainty Fisher's Separation Theorem and Tobin's Two-Fund Separation <u>Method</u> Primer on Dynamical Systems	<u>Core reading</u> - Copeland et al (Chapter 1) - Lecture notes (for both topics) <u>Further reading</u> - Fisher, I. (1930), The Theory of Interest. - Tobin, J. (1958), 'Liquidity preference as behavior towards risk', Review of Economic Studies 25(2).
	Week 2: Individual optimization under Risk Markowitz's Mean-Variance Frontier Deriving the efficient frontier for two assets, three assets and N-assets cases	<u>Core reading</u> - Copeland et al (Chapter 5) - Lecture notes (for both topics) <u>Further reading</u> - Markowitz, H. (1952), 'Portfolio selection', Journal of Finance 7(1).
	Week 3: The CAPM: From Individual Optimum to Market Equilibrium	<u>Core reading</u> - Lecture notes - Copeland et al (Chapter 6)
	Week 4: Review and Limits of the Static Framework - Rolls critique of CAPM - Limits of the fundamental assumption about the known distribution over future outcome	<u>Core reading</u> Lecture notes
CA 1: In-class test		
Block 2: Mainstream Dynamic	Week 5: Decisions under continuous time- Euler Method What does the agent do when today's decision alters tomorrow's opportunity set? - Myopia case - Euler-equation method	<u>Core reading</u> - Lecture notes <u>Further reading</u> - Samuelson, P. (1969), Review of Economics and Statistics 51(3). - Campbell & Viceira (2002), Strategic Asset Allocation, Chapter. 1.
	Week 6: Optimum versus Trajectory: Dynamics toolkit Stability of the trajectory Fundamental vs Bubble solutions	<u>Core readings</u> - Lecture notes - Shone, R. (2002), Economic Dynamics, Chapters. 2–4 - Raghavendra and Piiroinen (2023), Introduction to Economic Dynamics, Chapter 12
Mainstream Dynamic	Week 7: Merton's Continuous-Time Portfolio Problem - Wealth as the state variable and the current-value Hamiltonian for the consumption–portfolio problem	<u>Core reading</u> - Lecture notes <u>Further reading</u> - Campbell, J. Y., & Viceira, L. M. (2002). Strategic Asset Allocation, Chapter 5.

	Deriving the hedging demand from a moving (AR(1)) expected return	2. Merton, R. C. (1971), Journal of Economic Theory 3(4)
	Week 8: Review and Limits of Dynamic Portfolio problem <i>Every dynamic result this block produced is, at bottom, a Block 1 first-order condition either repeated at every date (Euler equation) or made instantaneous (Hamiltonian).</i>	
CA 2: In-class test		
Block 3: Heterodox Dynamic	Week 9: Keynes' Liquidity Preference as Portfolio choice theory - Kalecki's Principle of Increasing Risk- <i>Borrower's and lender's risk are endogenous and grow with the position</i> - Comparison with Merton's dynamic asset allocation model	<u>Core</u> - Lecture notes - Lavoie, M. (2014), Post-Keynesian Economics, Ch. 2. <u>Further readings</u> - Kalecki, M. (1937), 'The principle of increasing risk', <i>Economica</i> (New Series) 4(16). - Minsky, H. (1975), John Maynard Keynes, Chapter. 4
	Week 10: Financial Instability and Debt Dynamics - Minsky's two-price model- direct alternative to Block 1 static models - Steve Keen -debt financed investment dynamics leads to limit cycle and Hopf bifurcation	<u>Core readings</u> - Lecture notes - Minsky, H. (1992), Levy Economics Institute Working Paper No. 74. - Keen, S. (1995), <i>Journal of Post Keynesian Economics</i> 17(4)
	Week 11: Heterogeneous Agents, Reflexivity, and the Three Crises - Brock and Hommes model- fundamentalists vs chartists - Grossman and Stiglitz – Critique of Efficient Market Hypothesis	<u>Core readings</u> - Lecture notes - Brock, W. & Hommes, C. (1998), <i>Journal of Economic Dynamics and Control</i> 22(8–9) - Grossman, S. & Stiglitz, J. (1980), <i>American Economic Review</i> 70(3). <u>Further reading</u> Kirman, A. (1993), <i>Quarterly Journal of Economics</i> 108(1);
	Week 12: Cross-block review and synthesis	